# STATISTICS WORKSHEET-3

1 - b) Total Variation = Residual Variation + Regression Variation

2 - c) binomial

3 - a) 2

4- a) Type-I error

5 - c) Level of confidence

6- a) Decrease

7- b) Hypothesis

8- d) All of the mentioned

9- a) 0

10. What Is Bayes' Theorem?

The Bayes theorem is a mathematical formula for calculating conditional probability in probability and statistics. In other words, it's used to figure out how likely an event is based on its proximity to another.

P(A | B) = P(B | A)P(A) / P(B)

P(A | B) is the conditional probability of event A occurring, given that B is true.

P(B | A) is the conditional probability of event B occurring, given that A is true.

P(A) and P(B) are the probabilities of A and B occurring independently of one another.

11. What is z-score?

A z-score measures the distance between a data point and the mean using standard deviations. Z-scores can be positive or negative. The sign tells you whether the observation is above or below the mean. For example, a z-score of +2 indicates that the data point falls two standard deviations above the mean, while a -2 signifies it is two standard deviations below the mean. A z-score of zero equals the mean. Statisticians also refer to z-scores as standard scores

12. What is t-test?

A t-test is a statistical test that is used to compare the means of two groups. It is often used in hypothesis testing to determine whether a process or treatment has an effect on the population of interest, or whether two groups are different from one another.

13. What is percentile?

A percentile is a term that describes how a score compares to other scores from the same set. While there is no universal definition of percentile, it is commonly expressed as the percentage of values in a set of data scores that fall below a given value

14. What is ANOVA?

ANOVA means Analysis Of Variance

ANOVA is a technique used to test the difference between two or more means.

15. How can ANOVA help?

It provides the overall test of equality of group means. It can control the overall type I error rate (i.e. false positive finding) It is a parametric test so it is more powerful, if normality assumptions hold true